Action1 Entra ID Integration --- User Guide

The Action1 Entra ID integration synchronizes Microsoft Entra device information to Action1 managed endpoints. It brings Entra group membership and device ownership information into Action1 custom attributes, where that information can be used for endpoint filtering, grouping, reporting, and automation.

The integration reads data from Microsoft Entra ID through Microsoft Graph and updates matching endpoints in Action1. It does **not** make changes to Microsoft Entra ID.

How the integration works

During synchronization, the integration:

1. Reads devices from Microsoft Entra ID.
2. Reads each device's direct Entra group memberships.
3. Reads managed endpoints from the mapped Action1 Organizations.
4. Matches Entra devices to Action1 endpoints by device name.
5. Updates the configured Action1 custom attributes on uniquely matched endpoints.

By default, the integration populates:

- **Entra Groups** --- a comma-separated list of the device's direct Entra group memberships.
- **Entra Device Ownership** --- the Entra `deviceOwnership` value, such as `company` or `personal`.

Action1 credentials and the Action1 regional API endpoint are managed by Marketplace. No Action1 API URL, API key, or API secret is entered in this integration.

Before you begin

The following are required:

- A Microsoft Entra tenant.
- A Microsoft Entra application registration for the integration.
- The Entra Directory (tenant) ID.
- The application (client) ID.
- A client secret.
- The required Microsoft Graph **Application permissions**.
- Marketplace Action1 credentials that can discover Organizations, read managed endpoints, and update endpoint attributes.
- The required custom attributes in every Action1 Organization that will be synchronized.
- At least one Action1 Organization mapping.

Using a dedicated Microsoft Entra application registration for the Action1 integration is recommended.

Microsoft Graph permissions

The integration uses the OAuth 2.0 client credentials flow. It runs as an application rather than as a signed-in Microsoft user, so permissions must be configured as **Application permissions**, not Delegated permissions.

Configure:

Permission Purpose

`Device.Read.All` Read devices and device properties from Microsoft Entra ID

`GroupMember.Read.All` Read group memberships and the basic group information needed for synchronized group names

Both require administrator consent when used as Microsoft Graph application permissions. The integration is read-only on the Microsoft side and does not require Graph write permissions.

Microsoft documentation:

- Register an application with Microsoft Entra ID
- Microsoft Graph --- List devices
- Microsoft Graph --- List device memberships
- Microsoft Graph permissions reference
- Grant tenant-wide admin consent

> **Note:** The earlier standalone connector documentation used the broader `Directory.Read.All` permission. The Marketplace integration uses the more targeted permissions above for its current Microsoft Graph operations.

Step 1 --- Create the Microsoft Entra application

Sign in to the Microsoft Entra admin center with an account that can register applications and grant the required permissions.

1. Open **Entra ID** and go to **App registrations**.
2. Select **New registration**.
3. Enter a descriptive name, such as `Action1 Integration`.
4. For a normal single-tenant deployment, select the account type appropriate for the organization's tenant.
5. Select **Register**.

On the application's **Overview** page, copy:

- **Application (client) ID**
- **Directory (tenant) ID**

Add Microsoft Graph permissions

In the application registration:

1. Open **API permissions**.
2. Select **Add a permission**.
3. Select **Microsoft Graph**.
4. Select **Application permissions**.
5. Add `Device.Read.All`.
6. Add `GroupMember.Read.All`.
7. Select **Grant admin consent** for the organization.

Verify that the permissions show as granted before continuing.

Create a client secret

1. Open **Certificates & secrets**.
2. Select **New client secret**.
3. Enter a description such as `Action1 Marketplace`.
4. Select an expiration period that follows the organization's credential-rotation policy.
5. Create the secret.
6. Copy the **secret Value** immediately and store it securely.

Use the secret **Value**, not the secret ID, in Action1 Marketplace. When the secret expires, create a replacement and update the integration settings.

Step 2 --- Prepare Action1

Action1 access

Action1 API credentials are managed by Marketplace and are not entered again in the Entra ID integration.

The Marketplace credentials must be able to:

- discover Action1 Organizations;
- read managed endpoints;
- update managed endpoint attributes.

The integration only writes the configured custom attributes. It does not change endpoint names, endpoint status, or other standard Action1 endpoint properties.

Marketplace also supplies the correct regional Action1 API endpoint automatically. There is no Action1 region or API URL setting to configure.

Create the custom attributes

Before the first synchronization, create the destination custom attributes in **every Action1 Organization that will be mapped**.

The default names are:

- `Entra Groups`
- `Entra Device Ownership`

The names configured in Marketplace must match the Action1 custom attribute names exactly.

The integration does not create missing custom attributes automatically. If an attribute is missing in a mapped Organization, updates to affected endpoints can fail.

The names can be changed in Settings. Changing an attribute name later does not migrate or clear data previously written to the old attribute.

Step 3 --- Configure the integration

Open the Entra ID integration in Action1 Marketplace and go to **Settings**.

Entra Configuration

Setting Value

Entra Directory ID Directory (tenant) ID from the Entra application

Entra Application Client ID Application (client) ID

Entra Client Secret Client secret Value

Save the connection settings. The client secret is stored as a secret setting and is masked after saving.

Action1 Custom Attributes

Confirm the destination attribute names:

Setting Default

Entra Groups custom attribute `Entra Groups` name

Entra Device Ownership custom `Entra Device Ownership` attribute name

If different Action1 custom attribute names are used, enter those exact names and save the attribute settings.

Step 4 --- Test the Entra connection

Select **Test Connection**.

The test authenticates to Microsoft Entra, obtains a Microsoft Graph access token, and performs a basic Graph device read.

A successful test confirms that the integration can authenticate and perform its basic Microsoft Graph read operation.

Test Connection does **not** validate Action1 credentials, Action1 Organization mappings, Action1 endpoint permissions, custom-attribute existence, or a complete synchronization.

Step 5 --- Map Action1 Organizations

Open **Organization Mapping**.

The configured Entra tenant is the source. Mapping determines which Action1 Organizations receive synchronized Entra data.

1. Refresh the Action1 Organization list if necessary.
2. Add each Action1 Organization that should receive Entra data.
3. Save the mappings.

Multiple Action1 Organizations can be mapped to the configured Entra tenant. Each Action1 Organization can appear only once. Unmapped Organizations are ignored.

Mapping controls synchronization scope; it does not change device-matching rules.

Step 6 --- Run the first synchronization

Open **Sync** and select **Run Sync**.

A run:

1. Validates the Entra configuration, Marketplace Action1 connection, regional Action1 endpoint, and Organization mappings.
2. Authenticates to Action1 and Microsoft Graph.
3. Reads Entra devices and direct group memberships.
4. Reads managed endpoints from each mapped Action1 Organization.
5. Matches Entra devices to endpoints.
6. Builds custom-attribute updates.
7. Updates uniquely matched Action1 endpoints.
8. Records the result in Marketplace History.

Mapped Organizations are processed independently. A failure in one Organization does not automatically prevent other mapped Organizations from being processed.

The current integration applies safety limits of up to **50 endpoint updates per Organization job** and **200 endpoint updates per synchronization run**.

How devices are matched

Matching is based on device names:

- Microsoft Entra `displayName`
- Action1 `device_name`, with Action1 `name` as a fallback

Names are trimmed and compared case-insensitively.

```
Entra:   LAPTOP-01
Action1: laptop-01
Result:  Match
```

A short-name fallback also allows a fully qualified name to match its short hostname:

```
Entra:   LAPTOP-01.example.com
Action1: LAPTOP-01
Result:  Match
```

The integration does **not** use Entra device ID, serial number, MAC address, owner, enrollment profile, or compliance state for matching.

Duplicate and ambiguous matches

If multiple Action1 endpoints match the same normalized name, the match is ambiguous and is skipped.

If multiple Entra devices resolve to the same Action1 endpoint, endpoint-reuse protection also prevents that endpoint from being updated.

Unmatched Entra devices are skipped. Unmatched Action1 endpoints remain unchanged.

Entra group synchronization

For each Entra device, the integration reads its **direct** group memberships. Nested or transitive memberships are not expanded.

Group display names are written as a comma-separated value:

```
Finance, Remote Workers, Autopilot Devices
```

If a group display name is unavailable, the group ID is used. The integration preserves the order returned by Microsoft Graph and does not explicitly sort the list.

When a device has no groups

Empty values are not sent to Action1.

If an endpoint already contains an `Entra Groups` value and the Entra device later has no direct groups, the existing Action1 value is **not automatically cleared**.

Device ownership synchronization

The integration synchronizes the Microsoft Entra device `deviceOwnership` property.

For example:

```
Entra Device Ownership = company
```

This is the Entra device ownership property. It is **not** the name or email address of a registered owner or primary user. The integration does not query registered owners or primary users.

If Entra returns no ownership value, the field is omitted from the Action1 update and an existing Action1 value is not automatically cleared.

Example

Microsoft Entra:

```
Device: WIN-LAPTOP-01

Direct groups:
Finance
Remote Workers
Autopilot Devices

Device ownership:
company
```

Action1 managed endpoint:

```
WIN-LAPTOP-01
```

With the default custom attribute names, synchronization writes:

```
Entra Groups:
Finance, Remote Workers, Autopilot Devices
```

```
Entra Device Ownership:
company
```

The update occurs only when a unique device-name match is established.

Update behavior

For a uniquely matched endpoint, the integration sends non-empty Entra values to the configured Action1 custom attributes.

It does not first compare the new value with the current Action1 value, so unchanged non-empty values may be written again on later runs.

If a source value becomes empty, that field is omitted rather than cleared in Action1.

Endpoint-specific update failures are recorded while processing continues where possible.

Scheduling

Scheduling is managed by Action1 Marketplace and is **disabled by default**.

Available intervals:

- Every 12 hours
- Every 24 hours

Manual and scheduled runs use the same synchronization process. The Sync page prevents a new manual run from being started while another tracked synchronization is already running.

Sync status and History

The **Sync** page shows the current synchronization state, mapped Organization count, scheduler status, latest completed run, and result counters.

- **Synced** --- Action1 endpoints successfully updated.
- **Skipped** --- devices not updated, including unmatched, ambiguous, or endpoint-reuse cases.
- **Failed** --- matched endpoints whose update failed.

The **History** page displays up to the latest **30 runs**, including start time, result, synchronized count, skipped count, failed update count, and available run details.

Manual and scheduled runs use the same Marketplace History mechanism.

Troubleshooting

Test Connection fails with invalid credentials

Verify the **Entra Directory ID**, **Entra Application Client ID**, and **Entra Client Secret**. Make sure the client secret **Value** was used rather than the secret ID, and verify that it has not expired.

Test Connection reports insufficient permissions

Verify that `Device.Read.All` and `GroupMember.Read.All` were added as Microsoft Graph **Application permissions** and that administrator consent was granted.

Synchronization reports that no Organization is mapped

Open **Organization Mapping**, add at least one Action1 Organization, and save.

Devices are skipped

Check device names in Microsoft Entra and Action1. Matching uses Entra `displayName` and the Action1 endpoint name. Case differences do not matter and short-name/FQDN matching is supported. Duplicate or ambiguous matches are intentionally skipped.

Custom attribute updates fail

Verify that both configured custom attributes exist in every mapped Action1 Organization and that their names match Settings. Also verify that the Marketplace Action1 credentials can update endpoint attributes.

Group information is missing

The integration synchronizes **direct** device group memberships only. Nested memberships are not expanded. Also verify the Graph application permissions and admin consent.

An old group or ownership value remains in Action1

This can be expected. Empty Entra values are omitted from updates; the integration does not currently clear previously synchronized values.

One Organization fails but others continue

Mapped Organizations are processed independently. Review **History** for details about the affected run.

Setup checklist

Before enabling scheduled synchronization, verify:

- The Microsoft Entra application is registered.
- The Directory (tenant) ID and Application (client) ID have been recorded.
- A valid client secret has been created.

- `Device.Read.All` is configured as a Microsoft Graph Application permission.
- `GroupMember.Read.All` is configured as a Microsoft Graph Application permission.
- Administrator consent has been granted.
- Marketplace Action1 credentials can discover Organizations, read managed endpoints, and update endpoint attributes.
- `Entra Groups` and `Entra Device Ownership`, or the configured alternatives, exist in every mapped Action1 Organization.
- Entra connection settings are saved.
- **Test Connection** succeeds.
- Required Action1 Organizations are mapped.
- A manual synchronization has completed.
- Custom attribute values have been verified on at least one matched endpoint.
- History has been reviewed for skipped or failed devices.
- Scheduled synchronization is enabled if required.